Fostering Mobile Technology Value Proposition through User Participation: Results from two Industrial Case Studies

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Abstract

Mobile technology within corporate IT-Systems is a major and still increasing issue. Although the usage of mobile technologies for business processes is far behind expectations, still little research has been done in analyzing critical success factors for mobile technology acceptance and application after replacing a former paper-based process within the IT-Service domain. Therefore a mobile workforce solution for IT-Service technicians has been implemented and usability tested. Different extents of user centered design during the development and deployment of a mobile tool for service technicians have been observed in two industrial case studies.

Major results are (a) same archetypical business process metrics are affected after mobile tool deployment in different IT-Service companies, (b) major process improvements arise not from faster job processing but from faster and more accurate data access and transmission and (c) user participation in business process reengineering and mobile tool development fosters tool acceptance and reduces the return on investment time.

1 Introduction

Mobile information and communication technologies (ICT) offer a plethora of new value propositions like e.g. efficiency gains, mobile data access, improved data quality, etc. and promise to have a significant transformational impact on business processes, organizations, and supply chains [1][2][3]-[6]. Despite its potential contributions, enterprise adoption of mobile ICT has not been as widespread as initially anticipated [2][6][9]. Previous research indicated that successful adoption and implementation of any emerging ICT, such as mobile ICT, often requires fundamental changes [2] across an enterprise and its current business processes, organizational culture, and workflows [3][4]. Hence, in order to minimize organizational risks and maximize the potential benefits of mobile ICT, companies have to evaluate the value of mobility to their organization [7]. *Mobility* refers to the level of geographical independence of mobile workforce, and *reachability* refers to their ability to be contacted despite their mobility [8].

According to Basole [2] the potential value of mobile ICT in enterprises is tremendous. Enabling access to information and people anywhere and anytime, enhancing decision making capabilities through real-time information access, and creating a user-centric (mobile workforce) environment are only some examples of process changes through mobile ICT application in enterprises. Mobile ICT usage has become necessary through increasing customer and market demands of information readiness and flexibility of a company [20][21]. Furthermore, Basole [2] argues that the use of mobile ICT promises to make specific mobile process more efficient, effective and create a more personalized approach to information access, decision making, and communication than traditional wired environments [14]. Processes which are executed with mobile ICT are called *mobile-integrated business processes (MIBP).* A mobile-integrated business process is defined by the integration of mobile workforce via mobile technology into company business processes and IT-Systems (ERP-Systems) [4] (see Figure 1).
Mobile ICT differs from their non-mobile counter parts (e.g. desktop PCs, laptops) by factors like application context: the user of a mobile system is moving (travelling in public means of transport, walking or standing). The context of use is divers in e.g. lighting, noise and space. Whereas, technical limitations of mobile ICT are more and more solved, computing power, connectivity and battery lifetime are still limiting factors which have to be considered during development and deployment (e.g. optimize computation time for search functions in large data bases).

Value propositions of IT-Systems can be measured through *business metrics* and their contribution to decision support demands [24]. They can be a measure-

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ment for a company's economic success or failure e.g. cost/benefit, Balanced Score Card-Analysis or Return on Investment calculations (ROI) or Service Level Agreements (SLAs) [23]. *Business needs* are best served when business metrics – such as cost or revenue – are considered to measure business impact [24]. The *business impact* of certain information technology (IT) events (e.g. component faults) on business results must be captured and measured. Within the ITIL literature[1], impact is a measure of business criticality, often equal to the extent to which an incident leads to distortion of agreed SLAs (e.g. reachability of a service).

The remainder of this paper is structured as follows. Section 2 provides an overview of related work and illustrates business value propositions of mobile ICT application as well as how to overcome adoption barriers of mobile ICT. Section 3 describes the concept of mobile-integrated business processes and research model as well as two industry case studies. In section 4 the results are presented whereas section 5 discusses the findings. The paper concludes with a summary and provides an outlook to future research directions.

2 Related work

In most cases, enterprise adoption of mobile ICTs has been slower than originally anticipated [2][3][9]. This trend is often caused by technological limitations, security issues, and significant economic investments associated with the implementation of mobile ICT [2][9]. Despite these issues the main reason for the lack of enterprise adoption stems from the fact that current mobile applications fail to deliver a sufficiently compelling value proposition to corporate decision makers [20]. Value propositions of IT-Systems are measured through business metrics and their contribution to decision support demands.

Literature research for mobile business process comprised a number of international journals, books and conference papers. Despite the increasing relevance of the topic shown by numerous publications [2]-[3][22][25]-[27] in the area of mobile applications, a combined approach to investigate the impact of user participation on mobile ICT adoption and effects on business metrics is missing. The research approach within this paper is based upon qualitative, interpretative case study research [28]. This approach is especially appropriate for obtaining complex details and novel understandings about a specific phenomenon under investigation. Case study research is appropriate for examining practice-based problems, since it allows a researcher to capture the knowledge of practitioners [28]-[30]. In the course of this research we applied case study research to deepen the understanding of business processes impacts of mobile ICT application and investigate effects on business metrics. Moreover we investigated how different extents of user participation and user centered design during the development of mobile applications can foster tool acceptance and contribute to business process improvements.

Mobile business processes and the impacts on the development of mobile applications are discussed by Gebauer & Shaw [22]. They propose a framework where they define technology characteristics on the one hand and task characteristics on the other hand resulting in usage characteristics which have an impact on the actual usage of a mobile application. They define this impact through the usability variables efficiency (e.g. productivity: doing the things right) and effectiveness (e.g. performance: doing the right things).

According to Basole [2] the mobile enterprise readiness is defined by *seven dimensions* (technology, knowledge, process, leadership, resources, values and goals and end-users). Mobile enterprise readiness is defined as the preparedness and potential of adopting, implementing and using mobile ICT [2]. Prior to mobile enterprise readiness investigations, the business value proposition of mobile technology for a company/process, has to be defined. Within the case study companies under investigation this has been achieved by defining key performance indicators (KPIs) like: time to bill, service success rate, work load at the head office, paper handling time and Return on Investment. For a detailed description of the KPIs see chapter 4.

Basole [2] reflects in his Mobile Enterprise Adoption Framework the current state of mobile business processes together with the influencing factors of "mobile enterprise readiness" resulting into business values. Moreover he opposes the projected future state that should lead to a companys's future mobile business strategy. The discrepancy between those two states (current- and future state) leads to a potential for future mobile strategies that is represented in a gap between these two states. The business drivers can be classified as *strategic, economic*, and *social factors*. Within this research we focus on *economic factors* which represents one important issue of a companys' business metrics (e.g. response time, orders per day and mobile worker, driving time, billing time, etc.) which measure mobile ICT benefits and costs along a mobile ICT supported process.

The amount of increasing research (see: [8][22][25][27][31]-[33]) and the improved mobile technology (device, connectivity, performance) indicate that mobile business applications are gaining importance. Therefore we investigated which business

[1] ITIL: Information Technology Infrastructure Library: IT Infrastructure Library, "ITIL Service Delivery" and "ITIL Service Support", OG Commerce, UK. www.itil.org (Last Access: 12.01.2007).

metrics are affected by mobile tool usage and which factors serve as facilitators or inhibitors for mobile tool acceptance and usage.

The investigation of positive impacts of user involvement and user participation on system acceptance has been done extensively within the Inofrmation System (IS) literature over the last 30 years to name only a view e.g. [10]-[14]. Thereby, the terms *user participation* and *user involvement* are often used interchangeably within IS literature. However, in other disciplines, the concepts are accorded separately and have distinct meanings [15]. In order to address this anomaly, Barki and Hartwick [15] argue that the term *user participation* be utilized to refer to development-related activities and behaviours of users and their representatives during the development process, and that user involvement be used to refer to the subjective psychological state that reflects the level of importance and personal relevance of the information system to users. These researchers also argue that user participation is one of the more important antecedents, or causes, of user involvement - contingent on a number of factors which are said to influence the strength of the relationship.

User participation contributes to improve system quality by:

- Providing a more accurate and complete assessment of user information requirements [16][17].
- Providing expertise about the organization and the system to support expertise usually unavailable within the information system group [14].
- Avoiding development of unacceptable or unimportant features [17].
- Improving user understanding of the system [14][17].

User participation may lead to increased system acceptance by:

- Developing realistic expectations about system capabilities [18].
- Providing an arena for bargaining and conflict resolution about design issues [19].
- Leading to system ownership by users [17].
- Decreasing user resistance to change [14].
- Committing users to the system [14].

Whereas linking usability considerations and varying degress of user participation to the impact on business metrics have been investigate e.g. by [11]; investigations targeted at mobile applications in the IT-Service domain are missing according to the author's best knowledge.

Nielsen [11] estimates in his report "… spending about 10% of a project's budget on usability activities doubles usability". Within the above cited report metrics for usability evaluation of web sides have been gathered from 42 different cases. For the analysis projects that collected the same metric, both before and after the redesign of a web side, where needed in order to compare them and estimate the percentage improvement in usability.

A comparable procedure has been applied within the underlying case studies. Companies which collected business metrics of their mobile IT worforce, (e.g. response time, working- and driving time, etc.) before and after mobile ICT intetration have been selected and investigated during the transition process of former paper-based processes to mobile ICT supported processes.

2.1 Business value propositions of mobile ICT

The value propositions of mobility supported mobile workforce are numerous [2][32]. In this section we outline possible value propositions through information and resource access via mobile technology:

1. Relief of mobile worker from the desk: As information and resource access is possible anywhere, anytime (e.g. at the customer's site, while traveling via train). Task critical information can be retrieved timely which leads to timely answers and decisions.

2. Replacing paper-based processes: Paper-based processes are often error–prone and cause redundant work-steps (e.g. double data capturing on a paper form and transfer to DB later on): Higher data accuracy and integrity is reached through mobile ICT application which can be used for business intelligence applications.

3. Access to corporate resources: Instant access to people and resources leads to higher productivity [2] and shorter communication channels (see Figure 1).

4. Competitive Advantage and Customer Satisfaction: Mobile workers can react and respond faster to changing market conditions and customer demands than competitors.

5. Cost savings. Expensive computing equipment like laptops can be replaced with relatively cheaper handheld devices (e.g. PDAs).

6. Workload reduction in the head office: Through standardized communication channels, requests in the head office can be reduced as the mobile worker is able to retrieve job critical information independently e.g. mobile access to customer database.

Despite the numerous above mentioned value propositions of mobile ICT it is often difficult for decision makers to provide the required financial commitment. To facilitate this we present in the following section how a company can overcome adoption and transition barriers.

2.2 Overcoming mobile ICT adoption and transition barriers

Transition barriers can be classified as: *strategic/economic, technological, organizational* and *environmental* [3]. Whereas all the above mentioned barriers are important and need to be considered during the transformation from stationary, paper-based business processes to mobile-integrated business processes. In this contribution we focus on the *economic dimension* and how business processes can be supported with mobile ICT (technological dimension) as a complete consideration would go beyond the scope of this paper. In order to facilitate mobile ICT decisions it has to be assured that mobile-integrated business processes are align with the overall business strategy to avoid fragmented mobile ICT adoption and support enterprise future business objectives [32].

A further factor to overcome tool reluctance is the integration of end-users (User Centered Design - UCD) in the development process in order to increase tool usability [34][35] and thereby foster tool acceptance and usage after deployment. Mayhew [35] and Nielsen [34] show the importance of the usability variables "user experience", "user satisfaction" and "user acceptance". Therefore, we applied UCD within the development process of the implemented mobile tool and conducted a usability study in order to assure tool usability. Moreover, we investigated the degree of user participation in both case study companies to examine how user participation affects tool acceptance and the intension to use. Detailed results are presented in [3] as an in depth discussion of the usability evaluation results lays not within the scope of this paper. Table 1 provides an overview of the examined case studies.

3 Mobile-integrated business processes

This research investigates a business process representing an *archetypical workflow* for field technicians within the IT-Service domain [3]. The mobile business domain in this research applies to the mobile IT-Service process. The mobile service process includes all activities related to installation, service and maintenance jobs of a company's IT-Systems or services. In general a customer has a contract with a service providing company based on SLAs. Those SLAs specify for example how often a system has to go through maintenance checks or how long the maximum downtime of a system can last before the repair job has to be finished.

The process illustrated in Figure 1 implements the communication flow between a mobile worker (technician) and a dispatcher (head office) while using standard communication means like mobile phones. Prior to mobile tool support the mobile worker was not totally integrated into the mobile business process and a mobile phone and analogue devices like scratchpads were used for information exchange and communication within the process. The mobile worker had to contact the dispatcher every time he/she needs additional information which was not given during the first call from the dispatcher e.g. customer's location and SLA level, needed spare parts, earlier repair- and maintenance jobs for this customer, solution suggestions.

By modifying the process through the introduction of a mobile PDA based application (see Figure 1) we focus on the improvement of communication channels and autonomous information retrieval. A mobile-integrated business process is defined by the integration of mobile workforce via mobile technology into company business processes and IT-Systems [3]-[5]. This is realized via mobile tool support and effects mainly two parameters of the process:

Improving communication channels: Through the usage of mobile ICT communication channels can be improved by reducing the amount of necessary communication as well as the response times between the technician and the dispatcher (head office).

Improving information flow: We target at a standardization of the structure and content of transmitted information between the technician and the dispatcher via mobile devices (paper-based forms or desktop-based forms are replaced). We expect an improvement of the time to capture and process the information.

The depicted communication flows in Figure 1 reflects the whole information processed within the mobile business process after mobile tool integration. Within the process we distinguish between:

Working information: Describes the information transmitted in order to enable the mobile worker with all information necessary for conducting the process e.g. problem description, customer location, SLA-level.

Performance related information: This information is included in the workflow implemented in the mobile business process. Such information like e.g. execution- and driving time, used spare parts is useful for the controlling, planning and improvement of the process.

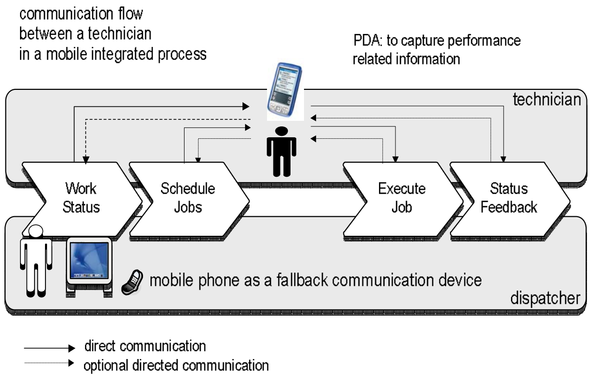

Figure 1: Integrated mobile worker

3.1 Research Model

In order to implement and evaluate the described communication and information improvement potentials we established a research model which enabled us to combine results from the prior usability evaluation and (two) industry case studies. We define the impact of user participation during the development process on business drivers. Within the research model UCD was applied and the frameworks of Gebauer and Shaw [22] as well as Basole [2] are combined to measure the impact of UCD, when implementing mobile applications, on business metrics. The results can serve for the development of future mobile business strategies (see Figure 2).

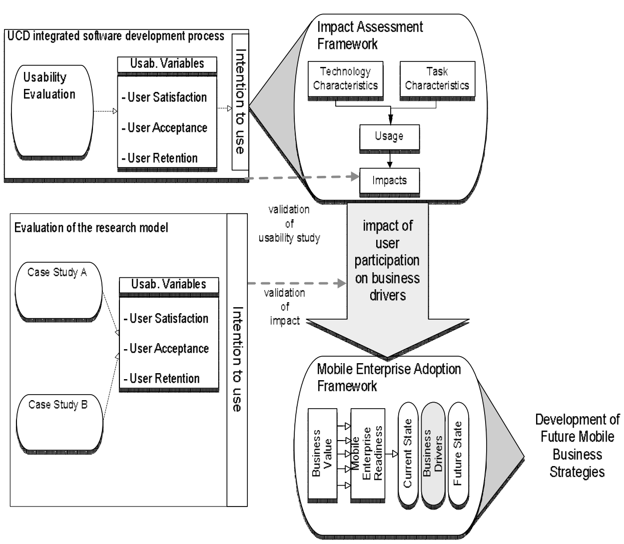

Figure 2: Framework - impact of user participation on business drivers

In order to base the study framework to reference values we implemented a mobile solution for IT-Service technicians (a detailed project description can be found in [37]). Within this project formative usability studies [36] were conducted and the results where applied iteratively during the software engineering process and further conducted two industry case studies. Moreover we applied the Impact Assessment Framework of Gebauer & Shaw [22] who defined the impact of usage parameters on the design of an application with respect to the impact on business metrics. The framework consists of four factors: technology, task, usage and impact according to which a mobile business application is investigated.

3.2 Case study description

As a common characteristic, the examined mobile business process, in both case studies, focus on technical customer service provision. Most important differences are service domain and company size [see Table 1]. The case studies were longitudinal studies in the IT-Service domain and were conducted in a time frame from 6 to 15 months. Cases have been selected considering the criteria of Yin [38]: relation to the research topic, rationale for multiple-case design and power to access information. Within the studies we applied user interviews as well as questionnaires and document analysis for data collection. The interviewees have been selected according to their functions in the project. Interviews have been transcripted and fed back to the interviewees in order to reduce possible errors and clarify misunderstandings. For data gathering CEO, CTO, project managers and end users of the mobile application have been questioned with semi-structured questionnaires (13 questions) in personal or via-phone interviews. Interviews lasted about 1.5 hours each.

In the Case Study A four persons have been interviewed. In the Case Study B three persons have been interviewed. Numbers of interviewees varied due to interviewee availability within the case study companies.

3.2.1 Case Study 1: IT-Service Provider

In this case study we investigated an Austrian IT-Service provider for hard- and software installation and maintenance. 40 IT service technicians have been equipped with mobile devices. The system was implemented in cooperation with an Application Service Provider (ASP). Field technicians use a browser-based solution on the mobile device in order to receive their jobs for the day which can be rescheduled via an implemented push-mail function. Main tasks of an IT-Service technician include repair, maintenance and installation of different hard- and software (e.g. repair of printers, reboot server, and exchange mother board). A major driver for MIBP introduction was to shorten the invoicing time, for which especially the records of working/driving time and used spare parts are essential.

3.2.2 Case Study 2: Toll Collection and Railway Maintenance Company

Within this case study we investigated an Austrian Toll Collection and Railway Maintenance Company. This company is responsible for the planning, financing, building and maintenance of the Austrian railway and highway system. The road network has a total length of 2,000 km at the time of study conduction (Status: Dec. 2006). 7 field service technicians have been equipped with a mobile tool (MDA III) to replace the former paper-based process. The mobile solution was also realized with the same ASP. That lead to the advantage that on the mobile device itself no sensitive data was stored but online connection was necessary for data transmission and retrieval which is available in all customer locations of both Case Study companies.

Table 1. Case study overview

Company	Service	# user	Architecture for MIBP	User Participation	Dispatching
A) IT-Service Provider*	Technical customer service	Small (40)	ASP solution, browser-enabled client	User participation in all development phases	Manually
B) Toll Collection and Railway Maintenance*	Technical customer service and trainings	Mini[2] (7)	ASP solution, browser-enabled client	Little user involvement; mainly in the device selection phase	Periodically execution according to geographic regions

4 Results

This section summarizes the investigated business process improvements in the case study companies. It was notable that the business process improvements in percent reached similar values in all case study companies e.g. concerning communication time, time to bill, etc. before and after mobile tool integration.

Major business process improvements could be realized through the reduction of the time to bill. Time to bill is the time needed to issue an invoice. As head office staff has faster (almost real time; directly after job completion or the latest at the end of a working day) job performance data access (working-, driving-time, used spare parts) the time to bill could be reduced. This is the time frame between job performance data is available in the head office and an invoice is send to the customer. The time to bill was reduced by about 50% in both case study companies. The Return on Investment was calculated as: *ROI = project costs / (benefit – current costs.)* ROI took 1.3 years in Case Study A and 1.6 years in Case Study B.

Results show that (a) same archetypical business process metrics are affected after mobile tool deployment in different IT-Service companies and (b) major process improvements arise not from faster job processing but from faster and more accurate data access and transmission and (c) user participation in business process reengineering and mobile tool development fosters tool acceptance and reduce return on investment time.In the Case Study A user participation was assured in all development phases of the mobile tool by applying UCD methods [34][35] whereas in the Case Study B users where only involved in the device selection phase. During the interviews users reported that they felt “observed” and “monitored” after mobile tool integration. Business process improvements and benefits have not been made transparent to the technicians. Tool reluctance was higher in Case Study B than in Case Study A. Initial tool reluctance also lead to a longer time for ROI in Case Study B. Beyond that all the above mentioned process improvements could be realized 3 to 6 months faster in Case Study Company A where user participation was assured during the whole implementation phase and process change. Process improvements could also be achieved in Case Study Company B but it took longer which had negative effects on the ROI. In both studies mobile ICT application realized business process improvements whereas a higher degree of user participation leads to faster process improvements and shorter ROI times.

* Usability evaluations of the mobile tool have been conducted in those companies.
[2] The number of field workers was only 7 therefore we categorized this company as a small company

Table 2 illustrates process improvements in the case study companies in minutes.

Table 2. Process improvement in minutes

	Without mobile Tool	Min	With mobile Tool	Min
Dispatching	The dispatcher had to call 3 technicians on average to ask for their current status, in order to find 1 available technician in the vicinity and assign calls.	15	Order dispatching is handled via automated or semi-automated systems. The dispatcher knows the current work status of technicians and can assign orders to them.	2
Job Execution	Inappropriate/Incomplete order information led to additional customer visits e.g. when needed spare parts where not available.	120	The technician receives detailed information about the order and can check if sufficient/appropriate spare parts are available.	30
	Spare part availability was not known exactly neither by the dispatcher nor the technician.	30	Access to spare part stocks and current availability e.g. spare parts in the car.	5
	Back office calls for information retrieval.	45	Information retrieval mainly via the mobile solution.	15
	Driving time per order	45	Driving time per order	30
Reporting	Transfer of paper-based reports.	30	Information is captured at the place of origin.	5

5 Discussion

In this contribution we argued that the adoption of mobile ICT is still a major issue within companies despite the high potential of business process improvement. Basole [2] described the existing gap between the assessment of the current mobile state of a company and future projected mobile strategies. This paper presents business metrics which are affected positively after mobile ICT application.

The results might help to bridge the gap described by Basole [2] by providing tangible value propositions of mobile ICT for decision makers. The empirical data gathered within this research serves as a facilitator to overcome adoption and transition barriers of mobile ICT. We identified generic business metrics like: billing time, paper handling time, communication effort and frequency etc. which are influenced by mobile tool integration into mobile business processes in the IT-Service domain.

Due to the limited amount of interviewees and industry case studies only in the IT-Service domain, generalizability of the results is arguable in varying application domains of mobile ICT. Despite these limitations this research provides a deeper insight and better understanding of the business impact of mobile tool integration into existing business processes. All case study companies stated that they are one-step ahead of the competition, due to increased information readiness, giving clients no reason to look for different service providers.

Mobile ICT is becoming an integral part of future IT-Service support processes. Businesses that are slow to integrate mobile ICT will experience competitors that are quicker to respond to customer requests and market demands.
The high dynamicity within the IT-Service domain creates the need for these businesses to work closely with mobile operators to look for innovative ways to improve communication and tie these technological advances to business needs. In the future mobile devices will be further enhanced and connection speeds improved (UMTS). The results for businesses that adapt early to these developments are: increased responsiveness and accountability as well as customer recognition, enhanced reputation and improved communication and information channels between the head office and the mobile workers and vice versa.

6 Conclusion and future work

We argued in this contribution that an understanding of the potential impact of mobile ICT on business processes is important for managers and decision makers in order to understand the potentials of mobile usiness processes. An assessment of an enterprise's preparedness, potential and willingness in combination with an understanding of the process changes will enable decision makers to formulate more objective judgments on why and when to adopt mobile ICT. During the execution of the described case studies we also experienced socio-technical impacts of mobile ICT and potential drawbacks of anytime, anywhere availability on the work-life balance. There is an argument that the ability to use dead time places greater pressure on the individual, making them accountable at times of the day that they would otherwise not be. However, it is up to the individual company to set these standards and ensure a suitable balance. Mobile working does not necessarily lead to longer hours; rather, it empowers the employee to manage his or her time autonomously. The investigation of socio-technical impacts of mobile ICT application will be part of our future work.

Future work will include the investigation of business impacts of mobile tool integration into existing company business processes in different industry domains. This should lead to the development of a unified framework for mobile-integrated business processes which can serve decision makers to choose business processes which are suitable for mobile tool support.

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7 References

[1] A. Kornak, J. Teutloff, and M. Welin-Berger, *Enterprise guide to gaining business value from mobile technologies*. Wiley, 2004.

[2] R. Basole, 2005, "Mobilizing the enterprise: A conceptual model of transformational value and enterprise readiness", in *Proceedings of the 2005 American Society for Engineering Management Conference*, Virginia

[3] K. Pousttchi and B. Thurnher "Understanding effects and determinants of mobile support tools: A usability-centered field study on it service technicians" in *ICMB '06: Proceedings of the International Conference on Mobile Business*, Washington, DC, USA: IEEE Computer Society, 2006a, pp. 8-10.

[4] K. Pousttchi and B. Thurnher, *Wireless Communication and Information*, Berlin, 2006b. Einsatz mobiler Technologien zur Unterstützung von Geschäftsprozessen, pp. 101–120.

[5] K. Pousttchi and B. Thurnher, "Adoption and impact of mobile-integrated business processes -comparison of existing frameworks and analysis of their generalization potential" in *Proceedings of Internationale Tagung Wirtschaftsinformatik,* 2007, pp. 273-290.

[6] W. B. Rouse, *Enterprise transformation: Understanding and enabling fundamental change.* Wiley, 2006.

[7] A. Hartman, J. G. Sifonis, and J. Kador, *Net Ready: Strategies for Success in the E-Conomy*. McGraw-Hill, 2000.

[8] S. Wang and W. Cheung, "E-business adoption by travel agencies: Prime candidates for mobile e-business," *International Journal of Electronic Commerce*, vol. 8, no. 3, pp. 43-63, 2004.

[9] R. Basole, "The value and impact of mobile information and communication technologies," *IFAC*, 2004.

[10] E. Kaasinen, *User acceptance of mobile services – value, ease of use, trust and ease of adoption.* VTT Publications, 2005.

[11] J. Nielsen, "Usability return on investment," *Nielsen Norman Group Report*, http://www.nngroup.com/reports/roi/, Report, 2003.

[12] P. Pedersen, L. Methlie, and H. Thorbjrnsen, "Understanding mobile commerce end-user adoption: a triangulation perspective and suggestions for an exploratory service evaluation framework," in *HICSS-35, Hawaii, US*, 2002, pp. 1-8.

[13] B. Ives and M. H. Olson, "User involvement and MIS success: A review of research," *Management Science*, vol. 30, no. 5, pp. 586-603, 1984.

[14] H. C. Lucas, "Systems quality, user reaction and the use of information systems," *Management Informatics*, vol. 3, no. 4, pp. 207-212, 1974.

[15] H. Barki and J. Hartwick, "Rethinking the concept of user involvement," *MIS Quarterly*, vol. 13, no. 1, pp. 53-63, 1989.

[16] D. Norton and F. W. McFarland, *The Information System Handbook*. Dow Jones-Irwin, Homewood, III, 1975, ch. Project Management, pp. 517-528.

[17] D. Robey and D. Farrow, "User involvement in information system development: A conflict model and empirical test," *Management Science*, vol. 1, pp. 73-85, 1982.

[18] H. L. Gibson, "Determining user involvement," *Journal of System Management*, pp. 20-22, 1977.

[19] P. G. W. Keen, "Information system and organizational change," *Communications of the ACM*, vol. 24, no. 1, pp. 24-33, 1982.

[20] M. Heck, "Mobilizing the enterprise: A conceptual model of transformational value and enterprise readiness," *Infoworld*, vol. 26, pp. 24-26, 2004.

[21] R. Kalakota and M. Robinson, *M-Business: The Race to Mobility.* McGraw-Hill Companies, 2001.

[22] J. Gebauer and M. Shaw, "Success factors and impacts of mobile business applications: Results from a mobile e-procurement study," *International Journal of Electronic Commerce*, vol. 8, no. 3, pp. 19-41, 2004.

[23] M. Hammer and J. A. Champy, *Reengineering the Corporation: A Manifesto for Business Revolution*, 1st ed. New York, NY, USA:Harpercollins, 5 1993.

[24] J. P. Sauvé, J. A. Moura, M. C. Sampaio, J. Jornada, and E. Radziuk, "An introductory overview and survey of business driven IT management. *International journal of mobile communications*, vol. 2, no. 4, pp. 354–367, 2004.

[25] U. Varshney, A. Malloy, P. Ahluwalia, and R. Jain, "Wireless in the enterprise: requirements, solutions and research directions," *International journal of mobile communications*, vol. 2, no. 4, pp. 354-367, 2004.

[26] E. Scornavacca, M. Prasad, and H. Lehmann, "Exploring the organisational impact and perceived benefits of wireless personal digital assistants in restaurants," *The International Journal of Mobile Communication*, vol. 4, no. 6, pp. 558-567, 2006.

[27] L. da Chen and R. Nath, "A framework for mobile business applications," *The International Journal of Mobile Communication*, vol. 2, no. 4, pp. 368-381, 2004.

[28] K. M. Eisenhardt, "Building theories from case study research," *Academy of Management Review*, vol. 14, no. 4, pp. 532-550, 1989.

[29] I. Benbasat, D. K. Goldstein, and M. Mead, "The case research strategy in studies of information systems," *MIS Quarterly.*, vol. 11, no. 3, pp. 369–386, 1987.

[30] J. W. Creswell, Research Design: *Qualitative, Quantitative, and Mixed Methods Approaches*, 2nd ed. Sage Publications, Inc, 7 2002.

[31] K. Fouskas, G. M. Giaglis, S. Karnouskos, P. E. Kourouthanassis, A. Pitsillides, and M. Stylianou, "A road-

map for research in mobile business," *International Journal of Mobile Communications*, vol. 3, no. 4, p. 350-373, 2005.
[32] R. C. Basole and W. B. Rouse, "Mobile Enterprise Readiness and Transformation*"*. Idea Group Inc. IGI. 2006.
[33] A. Gumpp and K. Pousttchi, "The "Mobility-M"-framework for application of mobile technology in business processes." in *GI Jahrestagung* (2), ser. LNI, A. B. Cremers, R. Manthey, P. Martini, and V. Steinhage, Eds., vol. 68. GI, 2005, pp. 523-527.
[34] J. Nielsen, "The usability engineering life cycle," *Computer*, vol. 25, no. 3, pp. 12–22, 1992.
[35] D. J. Mayhew, *The Usability Engineering Lifecycle: A Practitioner's Handbook for User Interface Design*. Morgan Kaufmann Publishers, 1999.
[36] A. Dix, J. Finlay, G. Abowd, and R. Beale: *Human-Computer Interaction*, 2nd edition, Pearson Education Limited, Essex, England, 1997.
[37] B. Thurnher, T. Kment, and J. Kjeldskov, "Long-term, large-scale usability evaluation methods - an empirical study," in M3 - Interdisciplinary Aspects on Digital Media & Education, M. Pohl, A. Holzinger, R. Motschnig, and C. Swertz, Eds. OCG, 2006, pp. 121-136.
[38] R. K. Yin, Case Study Research: Design and Methods. Sage Publications, Inc, 2002.

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